Process Simulator: an open, browser-based BPMN simulation environment with statistically controlled scenario comparison and simulation-based value stream mapping

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Abstract

What-if analysis with business process simulation usually compares a single simulated run per scenario, and value stream maps report only straight sequences of steps. We present Process Simulator, an open-source (Apache-2.0) discrete-event simulator for BPMN 2.0 that runs in the browser, as a Camunda Modeler plugin and as a command-line runner. It compares scenarios with replications, confidence intervals and common random numbers, and derives value stream figures (lead time, process cycle efficiency, value-added and non-value-added time, rework and waiting) that also cover exclusive and parallel branches and loops. We verify the engine against exact queueing results (M/M/1, M/M/c, M/G/1, tandem and Jackson networks): all 13 of 13 confidence intervals cover the exact value, worst relative error 0.86 %. A cross-check with Prosimos bounds the difference between the two engines to 5.9 %; on a model discovered with Simod from the BPI Challenge 2012 log the mean cycle times of both engines differ by -0.7 %. The value stream figures match closed-form results for sequences, choices, parallel splits and loops, where a linear map misstates process cycle efficiency. On a staffing problem, one run per scenario picks a wrong option in up to 43.8 % of the trials; sharing random numbers across options reduces this to 17.7 %, and with ten paired replications no wrong pick was observed in 100 trials (95 % upper bound 3.7 %). All results can be recomputed with one command.

Keywords: Business process simulation, BPMN, Discrete-event simulation, Common random numbers, Value stream mapping, Verification and validation, Open-source software

1. Introduction

A coffee-shop owner weighs three changes to her process: hiring a second barista, shifting the product mix to fifty-fifty, or both. She draws the process in BPMN, runs each option once in a simulation tool, and compares the reported mean cycle times. Whether she picks the right option depends on more than the model: in the decision experiment of this paper, one run per option with independent random numbers picks a wrong option in 21.0 % of the trials for this coffee shop and in 43.8 % on a harder office process (Section 4.5), although nothing about the model differs between the runs; only the random numbers do.

Business process simulation (BPS) estimates the effect of changes such as additional staff, shorter activities or different routing before they are implemented (Dumas et al., 2018; van der Aalst, 2015). Many BPM suites contain a simulator, and dedicated engines exist in research, notably BIMP (QBP) and Prosimos (López-Pintado & Dumas, 2022; López-Pintado, Dumas, & Berx, 2024), together with tools that discover simulation models from event logs (Camargo et al., 2020; Rozinat et al., 2009). Surveys of tool support report heterogeneous and often limited simulation capabilities (Laue & Müller, 2016; Pereira & Freitas, 2016; Rosenthal et al., 2021).

Two practical gaps motivated the tool presented here. First, what-if results are commonly read from a single run per scenario. The statistical remedies, independent replications, confidence intervals and common random numbers (CRN), are textbook simulation methodology (Kim & Nelson, 2001; Law, 2015; Nelson & Matejcik, 1995), but they are rarely part of the modelling workflow of BPMN tools. Second, value stream mapping (VSM) (Rother & Shook, 1999) is widely used to discuss lead time and waste, and simulation-based VSM is established in manufacturing (Lian & Van Landeghem, 2007; Solding & Gullander, 2009); process mining has been used to obtain value stream maps from logs (Knoll et al., 2019). A value stream map, however, is designed around a sequence of steps, whereas BPMN models contain exclusive choices, parallel branches and loops.

This paper describes Process Simulator and, mainly, its verification. We do not claim that BPMN simulation, simulation-based VSM or the combination of a model editor with a simulator are new. Two technical contributions accompany the environment: a formally defined, trace-based partition of sim-

ulated case lead time into value stream categories that remains well defined under non-linear control flow (Section 3.4), and a per-source random-stream scheme that keeps common random numbers synchronised across structurally different scenarios (Section 3.3). In full, the paper contributes:

1. modelling and simulation in one browser application without installation, and the same engine as a Camunda Modeler plugin and a command-line runner (Section 3);
2. scenario comparison with replications, Student-t confidence intervals and paired differences under CRN, where each source of randomness has its own stream so that changing one activity does not shift the random numbers of the others (Section 3.3);
3. value stream figures computed from the simulated time line of each case, which partition lead time also for choices, parallel branches and loops (Section 3.4);
4. an automated evaluation: exact queueing results, a cross-check with Prosimos, a model discovered from a public event log, closed-form VSM cases, decision quality of single runs versus replications, and run time (Section 4);
5. a reproducibility package under Apache-2.0 that recomputes every number in this paper with one command.

2. Related work

BPS tools and engines. BPS has a long tradition (van der Aalst, 2015); the BPSim standard defines simulation parameters for BPMN models (Laue & Müller, 2016). Pereira and Freitas (2016) and Rosenthal et al. (2021) compare simulation support in BPM tools and observe large differences in the supported parameters and outputs. Prosimos is an open-source Python engine with differentiated resources, i.e. individual calendars and performance per resource (López-Pintado & Dumas, 2022); it is the engine behind Simod, which discovers simulation models from event logs (Camargo et al., 2020; López-Pintado et al., 2024). Our engine is simpler in this respect: resources form roles with a shared calendar and a single duration distribution per activity.

Quality of simulation models. Discovered models are usually assessed by comparing simulated and real logs (Chapela-Campa et al., 2023, 2025) or by

their usefulness for downstream tasks (Özdemir, Kirchdorfer, Amiri Elyasi, van der Aa, & Stuckenschmidt, 2025). Blümel, Kirchdorfer, Rehse, Kampik, and van der Aa (2026) identify systematic errors in the evaluation of data-driven simulation models that can distort the assessment of model quality, even for the same process and model. These works concern model discovery; we use a discovered model only to check that our engine reproduces a second engine on realistic input (Section 4.3).

Comparing alternatives. Selecting the best of several simulated systems is the subject of ranking and selection (Bechhofer, 1954; Kim & Nelson, 2001); CRN reduce the variance of differences between systems (Law, 2015; Nelson & Matejcik, 1995). Independent streams per source are a standard way to keep random numbers synchronised across scenarios (L'Ecuyer et al., 2002). Testbeds such as SimOpt collect simulation-optimisation problems (Eckman et al., 2023; Pasupathy & Henderson, 2006). We implement replications, confidence intervals and CRN; we do not implement a ranking-and-selection procedure.

Value stream mapping. Simulation-based VSM has been proposed for manufacturing systems to add dynamics to the static map (Lian & Van Landeghem, 2007; Solding & Gullander, 2009), and multidimensional process mining has been used to build value stream maps for internal logistics (Knoll et al., 2019). Our VSM view defines a mutually exclusive partition of the simulated case time line (Section 3.4), so the figures remain well defined and conservative under choices, parallelism and rework in BPMN models.

3. The tool

3.1. Architecture

Figure 1 shows the application in simulate mode on the coffee-shop example shipped with the tool: the model carries annotations such as durations and branch probabilities, scenario tabs sit above the diagram, and the inspector on the right holds the demand, resource and activity parameters of the selected scenario.

Process Simulator is a TypeScript application. The modeller is bpmn-js; simulation parameters are stored inside the BPMN file as an extension element, so a model and its scenarios travel as one file. The discrete-event engine has no browser dependencies and runs in a Web Worker in the browser,

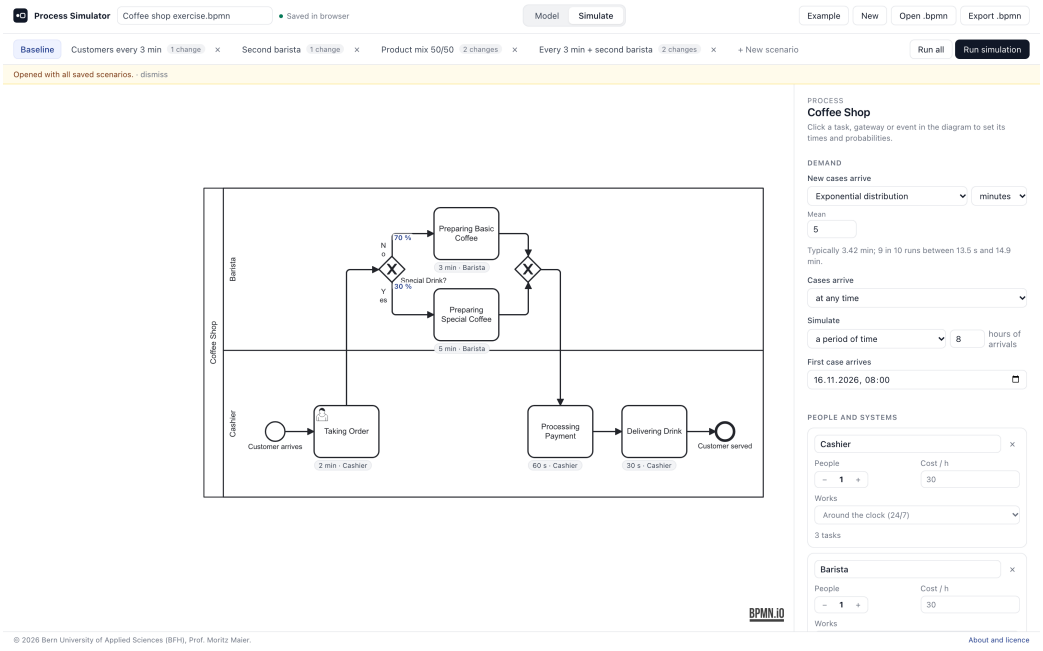


Figure 1: Process Simulator in the browser on the shipped coffee-shop example. Above the diagram, tabs switch between the baseline and the stored what-if scenarios; activity annotations show duration and role; the inspector edits the arrival process, resource pools and per-activity parameters of the selected scenario.

in the Camunda Modeler plugin and in Node.js through the command-line runner `procsim`. The runner reads a BPMN file, runs one or all stored scenarios, or a scenario given as JSON, for a given number of replications, and writes one JSON line per run with the simulator version, truncated SHA-256 hashes of the model and the scenario, the seed, the replication index and the random-number mode. It also exports scenarios to Prosimos parameters and imports Prosimos parameters, e.g. from Simod.

3.2. Simulation model

A scenario specifies the arrival process (inter-arrival distribution and optionally a calendar of the hours during which cases arrive), the number of cases or a time horizon, roles with a number of resources, an hourly rate and a weekly working-hours calendar, and for each activity a duration distribution (fixed, exponential, uniform, triangular, normal, lognormal or gamma), a role, an optional cost per run, revenue, queue limit, repetition probability

and value class (value-added VA, business-value-added BVA or non-value-added NVA). Branch probabilities are set on the outgoing flows of exclusive and inclusive gateways. The engine executes BPMN token semantics for sequence flows, exclusive, inclusive, parallel, complex and event-based gateways, embedded sub-processes, multi-instance activities, intermediate timer and message events and boundary events. Activities wait in a first-come-first-served queue for a free resource of their role; work outside the calendar is paused and resumed at the next opening. Results include cycle, waiting and processing times (mean and percentiles), throughput, work in progress, utilisation per role, cost and revenue, per-activity statistics and a simulated event log.

3.3. Statistical scenario comparison

Each scenario is simulated in R independent replications (default ten); replication i of every scenario uses the same seed s_i. In the default *keyed* mode, every random draw comes from a stream identified by its source (arrival, activity, gateway, boundary event or repetition), the case number and the index of the invocation: the n-th time a source is asked for randomness within a case, a small-state generator (sfc32) is seeded with a 32-bit hash of the replication seed, the source key, the case and n. A change to one activity therefore leaves the arrival times, branch decisions and durations of all other sources unchanged, and loops keep their draws synchronised per visit. Per-source streams are our implementation choice for CRN, not a requirement of CRN itself (Nelson & Matejcik, 1995): their purpose is synchronisation of corresponding stochastic events across scenarios. Synchronisation can still degrade when a structural change alters how often a source is visited within a case (for example a new loop), because subsequent invocations of that source shift index; draws of untouched sources remain paired. Also, hash-seeded streams are deterministic by construction; their statistical independence is assumed, not proven. A legacy mode with a single stream for everything is kept for comparison (Section 4.5).

Arrivals stop when the requested number of cases has arrived or a set time horizon is reached; the run then continues until every arrived case has completed, so the statistics are not restricted to cases that happen to finish early. Cases can remain unfinished only through a deadlock in the model or when a safety cap on the number of events stops the run; the output reports this, and it did not occur in the experiments reported here. Warm-up is

specified as simulated time; cases that arrive before the end of the warm-up are excluded from all statistics.

For each output measure the tool reports the mean over replications with a 95 % Student-t confidence interval. Scenarios are compared with the baseline by the paired differences $d_i = Y_i^{\text{scenario}} - Y_i^{\text{baseline}}$; the difference is reported with its confidence interval and marked as not significant when the interval contains zero. These intervals support pairwise estimation; they do not provide simultaneous error control across several alternatives, nor a guaranteed selection of the best one (Section 4.5 quantifies the consequences). The same intervals appear in the app, in the PDF and Excel reports and in the JSON output.

3.4. Simulation-based value stream mapping

For every completed case the engine records the intervals in which an activity of the case was being processed, together with the value class of the activity and whether the execution was a repetition (rework). The lead time of the case is then partitioned by sweeping its time line. Formally, let T_c be the lead time of case c and $I_{c,k}(t)$ indicate whether at least one activity of category k of case c is in process at time t. With the precedence order VA $\prec$ BVA $\prec$ rework $\prec$ NVA, each instant is assigned to

$$g_c(t) = \min_{\prec} \{k : I_{c,k}(t) = 1\}, \tag{1}$$

and to waiting when no activity is in process. The time attributed to category k is

$$T_{c,k} = \int_0^{T_c} \mathbf{1}[g_c(t) = k]\, dt, \qquad \text{so that} \qquad \sum_k T_{c,k} = T_c. \tag{2}$$

The partition is unique given the chosen precedence order; the order itself is a modelling decision, not dictated by Lean theory. With parallel branches, an instant in which a VA activity and an NVA activity run simultaneously counts as VA: the NVA work remains visible in the per-activity statistics but does not extend the lead time.

Two consequences deserve emphasis. First, this PCE measures the share of elapsed case time covered by value-adding work, which differs from the work-content share of VA effort: when VA and NVA work overlap, all elapsed time counts as VA while resources also consume NVA effort (Section 4.4 includes a case built to expose this). The per-activity statistics give the complementary work-content measure, the VA share of all executed work;

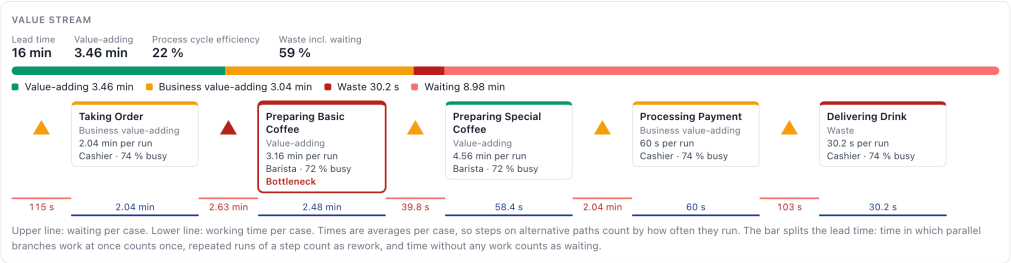


Figure 2: Simulation-based value stream map of the coffee-shop example. The bar partitions the mean lead time of the run shown (16 min) into value-adding, business-value-adding, waste and waiting; the bottleneck activity is marked. Over ten replications the baseline averages 13.5 min and a PCE of 27 %. Below, the upper line gives waiting and the lower line working time per activity. A linear map would add the durations of both coffee branches and report a PCE of 70 % instead of the simulated 22 %.

Section 4.4 reports both. Second, the figures depend on the value classes assigned by the modeller. The value stream view reports the mean of each category, the process cycle efficiency PCE = VA/lead time and the per-activity contributions; it also shows the classical linear figure, the sum of the activity times along the model, to make the difference visible. Figure 2 shows the view on the simulated coffee-shop baseline.

4. Evaluation

All experiments use the command-line runner of version 1.0.1 and are scripted in the paper repository; `make` recomputes data, tables and figures. Unless stated otherwise, 95 % confidence intervals are Student-t intervals over independent replications. The order of the experiments follows the question that opened this paper: Sections 4.1–4.4 establish that the engine and its value stream figures can be trusted, Section 4.5 measures how often the single-run shortcut fails, and Sections 4.6–4.7 place run time and feature coverage in context. We distinguish *verification* (does the engine implement the intended model correctly?), *cross-engine consistency* (does it agree with an independent implementation?) and *model fidelity* (does the simulated model reproduce a real process?); the experiments address the first two, not the third. Appendix A shows the test models.

Table 1: V1: simulated versus exact mean times in seconds; 13 verification quantities across the tested models.

Model	Quantity	Exact	Simulated (95 % CI)	Error	In CI
M/M/1, ρ=0.5	mean wait	30.0	30.0 ± 0.2	-0.14 %	yes
M/M/1, ρ=0.8	mean wait	192.0	191.4 ± 2.5	-0.33 %	yes
M/M/1, ρ=0.9	mean wait	486.0	483.8 ± 13.0	-0.46 %	yes
M/M/3, ρ=0.8	mean wait	155.3	154.9 ± 2.5	-0.24 %	yes
M/M/10, ρ=0.9	mean wait	361.1	360.7 ± 13.1	-0.11 %	yes
M/D/1, ρ=0.8	mean wait	96.0	95.9 ± 0.9	-0.10 %	yes
M/U/1, ρ=0.8	mean wait	104.0	104.1 ± 0.9	+0.09 %	yes
M/G/1 lognormal cv=1.5, ρ=0.8	mean wait	312.0	311.4 ± 7.6	-0.19 %	yes
M/G/1 gamma cv=0.5, ρ=0.8	mean wait	120.0	119.8 ± 1.1	-0.15 %	yes
Tandem M/M/1 to M/M/1	mean wait station 1	80.0	79.8 ± 0.6	-0.25 %	yes
Tandem M/M/1 to M/M/1	mean wait station 2	192.0	190.4 ± 2.7	-0.86 %	yes
Jackson feedback, p=0.25	mean wait per visit	144.0	143.9 ± 1.7	-0.06 %	yes
Jackson feedback, p=0.25	mean time in system	240.0	239.8 ± 2.3	-0.07 %	yes

4.1. V1: does the engine reproduce exact queueing results?

We built BPMN models whose steady-state behaviour is known analytically: M/M/1 at utilisation $\rho \in \{0.5, 0.8, 0.9\}$, M/M/3 and M/M/10, M/D/1, M/U/1 and M/G/1 with lognormal and gamma service (Pollaczek–Khinchine), a tandem of two M/M/1 stations and a Jackson network with feedback (Jackson, 1957). Each case ran 30 replications of 100 000 cases with a warm-up of 2000 mean inter-arrival times. Table 1 compares the simulated mean waiting time (or cycle time) with the exact value. All 13 of 13 intervals contain the exact value, and the largest relative error is 0.86 %. With correct marginal coverage about 12.4 of 13 intervals would be expected to cover the truth, so the observed result is consistent with nominal coverage without proving it. Little's law $L = \lambda W$ (Little, 1961), computed from the reported work in progress, throughput and cycle time, holds within 0.03 %.

4.2. V2: do the engines agree on shared semantics?

Five models (tandem, M/M/3, exclusive choice, parallel split and rework loop) were exported with the built-in converter and simulated by Prosimos 2.1.0 and by Process Simulator, 20 replications of 50 000 cases each. The two engines use different random-number generators, so we compare means. Table 2 lists the relative difference with a Welch confidence interval. The intervals contain zero for 13 of 13 quantities, but a confidence interval covering zero does not establish equivalence. As a descriptive yardstick, not a pre-specified equivalence test, we report how many 95 % intervals lie entirely within ±5 %: 11 of 13 do, and the widest admits differences up to 5.9 %

Table 2: V2: Process Simulator versus Prosimos, mean times in seconds and relative difference with 95 % confidence interval.

Model	Quantity	This tool	Prosimos	Difference (95 % CI)
Sequence, two M/M/1 stations	mean cycle time	360.6	357.7	+0.8 % [-1.2, +2.8]
Sequence, two M/M/1 stations	mean wait t1	80.1	79.6	+0.7 % [-1.4, +2.7]
Sequence, two M/M/1 stations	mean wait t2	192.5	190.1	+1.3 % [-2.0, +4.5]
One activity, three servers	mean cycle time	301.3	298.7	+0.9 % [-1.3, +3.1]
One activity, three servers	mean wait t1	157.3	154.5	+1.8 % [-2.2, +5.9]
Exclusive choice 30/70, shared role	mean cycle time	289.2	292.1	-1.0 % [-3.7, +1.8]
Exclusive choice 30/70, shared role	mean wait a	236.5	239.7	-1.3 % [-4.7, +2.1]
Exclusive choice 30/70, shared role	mean wait b	237.4	240.2	-1.2 % [-4.5, +2.1]
Parallel split, two roles	mean cycle time	246.6	242.8	+1.6 % [-0.4, +3.5]
Parallel split, two roles	mean wait a	136.0	133.4	+1.9 % [-0.9, +4.7]
Parallel split, two roles	mean wait b	141.8	138.3	+2.5 % [-0.3, +5.3]
Rework loop p = 0.25	mean cycle time	243.8	239.8	+1.7 % [-0.6, +4.0]
Rework loop p = 0.25	mean wait work	146.8	143.8	+2.1 % [-0.7, +5.0]

(the waiting time of the three-server model). Eleven of the thirteen point estimates of Process Simulator are slightly higher (by 0.7–2.5 %), so a small systematic difference cannot be excluded at this sample size; it is of the order of the interval half-widths. The data bound the difference between the engines to a few percent on the shared semantics; they do not show that either engine is correct. Features without a Prosimos counterpart, such as queue limits, revenue or boundary events, are listed by the exporter and were not used. One larger example model, the coffee shop shipped with the tool, could not be run by Prosimos (an internal error in its control-flow handling), which we report as a limitation of the cross-check.

4.3. V3: does the engine stay consistent on a discovered model?

To exercise the engine on realistic input we used the BPI Challenge 2012 log (van Dongen, 2012), restricted to the work items (activities starting with `W_`) with start and completion events, i.e. 9 657 cases. Simod 5.2.1 (Camargo et al., 2020) discovered a BPMN model with branching probabilities, resource profiles, calendars and duration distributions. Prosimos simulated this model directly; Process Simulator imported the Prosimos parameters. Because Process Simulator has one role and one duration per activity, the importer pools roles that share activities and merges the resource-specific durations of an activity into a gamma distribution with the same mean and variance; it reports 8 such approximations. Table 3 compares both engines with the log. The mean cycle times of the two engines differ by −0.7 %;

Table 3: V3: BPI Challenge 2012 work items, log versus simulation of the discovered model, ten replications each (mean ± half-width of the 95 % confidence interval).

Measure	Log	Prosimos	This tool
mean cycle time (h)	252.02	3.42 ± 0.05	3.40 ± 0.03
work items per case	7.39	17.86 ± 0.07	17.80 ± 0.06

since Prosimos simulated the original model and Process Simulator its imported approximation, this figure bounds the combined effect of the import approximations and any remaining semantic difference. Both engines also produce about 17.8 work items per case against 7.4 in the log, so the discovered control flow repeats activities far more often than the real process; the two engines agree with each other on this as well. The cycle times of both engines are far below the cycle time in the log, because the discovered model does not represent the time cases spend waiting outside the recorded work items; this is a property of the discovered model, not of either engine, and illustrates the threats discussed by Blümel et al. (2026). This experiment is therefore a consistency test between two implementations on realistic input, not a validation of predictive fidelity against the log.

4.4. V4: are the value stream figures correct for non-linear processes?

Seven models with closed-form lead time and partition were simulated with 30 replications: a sequence of three activities, an exclusive choice (30/70), a parallel split, a parallel split with value-adding and non-value-adding work of equal duration, a composite of a choice, a parallel pair and a rework loop, a rework loop with repetition probability 0.5 and a tandem with queues. Table 4 compares lead time and PCE with the exact values and with the PCE of a linear value stream map that adds the activity times along the model. The simulated figures agree with the exact ones within sampling error, and in every case the categories add up to the lead time (largest deviation 2×10^{-14} s). The linear map overstates PCE for the choice (because it adds both branches), understates it for the parallel split (because it adds parallel work) and ignores rework. The equal-duration case makes the semantics of the partition explicit: the simulated PCE is 1.0 although half of the work content is non-value-adding, because every instant of elapsed time carries VA work; the NVA effort is reported in the activity statistics but does not extend the lead time. Table 4 also gives the work-content share, which is 0.5

Table 4: V4: lead time (s), PCE, rework time (s) and VA share of executed work: exact, simulated and from a linear value stream map.

Case	Lead time		PCE			Rework	VA work share	
	exact	tool	exact	tool	linear map	tool	exact	tool
Sequence	60.0	60.0	0.167	0.167	0.167	0.0	0.167	0.167
Exclusive choice (30/70)	58.0	57.9	0.517	0.515	0.714	0.0	0.517	0.515
Parallel split	50.0	50.0	1.000	1.000	0.625	0.0	0.625	0.625
Parallel VA + NVA, equal duration	50.0	50.0	1.000	1.000	0.500	0.0	0.500	0.500
Choice + parallel + rework (composite)	58.4	58.4	0.633	0.633	0.550	6.4	0.654	0.654
Rework loop (p = 0.5)	40.0	39.9	0.500	0.502	1.000	19.9	1.000	1.000
Sequence with queues (M/M/1 tandem)	360.0	358.1	0.111	0.112	–	0.0	0.455	0.455

in this case: the two measures answer different questions and should be read together.

4.5. V5: single runs versus replications

We return to the question that opened this paper: which option is best, and how often does one run per option answer it wrongly? We took two decision problems: the coffee-shop example (four alternatives to the baseline) and an office process with a baseline and four staffing and duration options (Figure A.2 and Table A.1 give the office process and its scenarios, Figure A.3 and Table A.2 the coffee shop). The reference ranking comes from an independent high-precision run: 2 000 replications per option in a seed block disjoint from the trials, so no replication contributes to both the reference and the evaluated decisions. In both problems the best option leads the runner-up by a clearly resolved margin (21.8 % and 9.8 % of mean cycle time). We then repeated, 1 000 times, the decision "pick the option with the lowest mean cycle time", sweeping the budget from one run per option to groups of up to 50 paired replications; Table 5 reports the share of correct selections with Wilson intervals. With one run per option the answer is wrong in 21.0 % of trials on the coffee shop and 43.8 % on the harder office problem when seeds are independent; on the office process, sharing random numbers reduces this to 17.7 % with per-source streams and 23.9 % with a single shared stream, and on the coffee shop a single run with per-source streams is almost always right. Five paired replications already decide correctly in 99.5 % of office trials; with ten replications no wrong pick was observed (0 of 100, 95 % upper bound 3.7 %); the larger budgets rest on fewer trials and are correspondingly less precise. The paired confidence intervals of the tool never declared a wrong direction and returned "not significant" in at most

Table 5: V5: share of 1 000 independent decision trials in which the truly best option was chosen, by random-number mode and replication budget per option; brackets give the 95 % Wilson interval. The reference ranking uses a disjoint set of 2 000 replications per option.

Problem	Best gap	One run per option			Paired replications per option				
		one stream	per-source	indep. seeds	2	5	10	25	50
Coffee shop	21.8 %	99.4 [98.7, 99.7]	99.9 [99.4, 100.0]	79.0 [76.4, 81.4]	100.0 [99.2, 100.0]	100.0 [98.1, 100.0]	100.0 [96.3, 100.0]	100.0 [91.2, 100.0]	100.0 [83.9, 100.0]
Office process	9.8 %	76.1 [73.4, 78.6]	82.3 [79.8, 84.5]	56.2 [53.1, 59.2]	89.8 [86.8, 92.2]	99.5 [97.2, 99.9]	100.0 [96.3, 100.0]	100.0 [91.2, 100.0]	100.0 [83.9, 100.0]

Table 6: V5: variance of the difference to the baseline in mean cycle time (s^2) for one run per option.

Problem	Option	True difference (s)	Variance of the difference		
			independent	one stream	per source
Coffee shop	Customers every 3 min	+951.5	327173	279529	254513
Coffee shop	Second barista	-157.2	28334	11993	10595
Coffee shop	Product mix 50/50	+183.1	102702	41657	30029
Coffee shop	Every 3 min + second barista	+157.8	28119	20395	18007
Office process	Second person at step 1	-179.2	17446	6128	3190
Office process	Step 2 10 % faster	-116.4	16098	4298	1822
Office process	Step 3 11 % faster	-58.2	19834	4660	231
Office process	Second person at step 3	-124.6	18939	5545	649

4.2 % of the comparisons with the baseline. Table 6 shows why: per-source streams reduce the variance of differences by up to a factor of 20 compared with a single stream. For the coffee-shop owner of the introduction the lesson is twofold: comparing the options under the same random numbers already removes most wrong picks, because every option then sees the same customers and the same durations, and a handful of paired replications removes the rest; the confidence intervals then say when the evidence is insufficient instead of hiding it.

4.6. V6: how does run time scale?

Run time measures the implementation on a given machine, not the tool as such: it depends on the processor, the JavaScript engine and the load on the machine, whereas the simulated results do not. We therefore ran the same workloads on one machine (Apple M3, 8 threads) with the command-line runner (Node.js v20.18.3) and with the engine in a Web Worker, as in the web application, in headless Chromium and Firefox. Each point is the median of 5 replications and times the simulation only, not loading the model; Firefox reports these times in whole milliseconds. The engine's safety cap on the number of events was raised via the runner's `-max-events` option so that every requested case ran to completion. For the same seeds all three run-

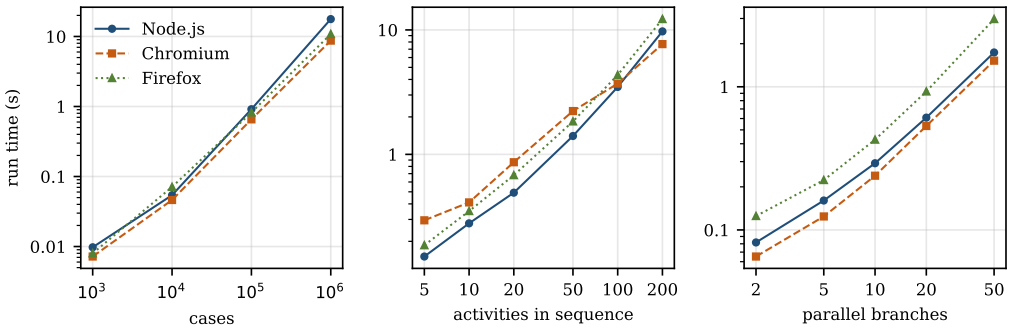


Figure 3: V6: run time by number of cases, sequence length and number of parallel branches, in Node.js and in a Web Worker in Chromium and Firefox on the same machine (logarithmic axes; median of 5 replications).

times produced identical results (0 differences in 30 comparisons). Figure 3 shows that run time grows faster than linearly with the number of cases beyond about 10^5 (throughput falls from 185 000 to 56 000 cases per second in Node.js), consistent with the per-case records the engine keeps for its statistics; one million cases of a one-activity process take 17.7 s in Node.js, 8.7 s in Chromium and 10.8 s in Firefox. Run time grows slightly faster than linearly with the number of activities or parallel branches per case, since every token event is a heap operation on a larger event list and more per-activity statistics are kept. Relative to Node.js, the browsers took between 0.5 and 2.0 times as long; on several workloads the browser engines were faster than the command-line runner. Within a workload, the range of the five replications was typically 25 % of the median (median over workloads), with isolated slow replications on short workloads, which the median absorbs; rerunning the experiment shifted the medians by up to about a fifth, so the absolute times are indicative. On other hardware the absolute times will differ, but the scaling is a property of the engine.

4.7. V7: how does the tool compare with existing ones?

Table 7 positions Process Simulator against two openly documented research tools that serve the same purpose, BIMP (the online front end of the QBP simulator) and Prosimos with Simod. Browser access alone is not the differentiator: both other tools can be reached through a web interface. What distinguishes Process Simulator is the integration of modelling, simulation, replicated comparison and the value stream view in one browser application,

and its particular statistical and VSM features. The entries for Prosimos were checked against the source code of release 2.1.0, the latest at the time of writing (command-line options, simulation API and random-number generation), and against its web service; the entries for BIMP come from its public documentation and its open-source user interface, as the QBP engine itself is not published. All sources were consulted in November 2026. "Not documented" means that the feature appears neither in the documentation nor in the inspected code, not that it is absent. Commercial suites with simulation modules were not included because their feature sets could not be checked from public sources in a comparable way; surveys of such tools are available (Pereira & Freitas, 2016; Rosenthal et al., 2021).

Table 7: V7: feature comparison based on public documentation and, for Prosimos, the source code of release 2.1.0.

Feature	Process Simulator	BIMP (QBP)	Prosimos / Simod
Runs in the browser, no installation	yes	yes (upload of a BPMN file)	engine in Python; open-source web application available
Integrated BPMN editor	yes (bpmn-js, Camunda Modeler plugin)	no, external modeller	no, external modeller
Command-line / batch use	yes	not documented	yes
Discovery from event logs	no (imports Simod output)	no	yes (Simod) (Camargo et al., 2020)
Differentiated resources (individual calendars and durations)	no, roles	no	yes (López-Pintado & Dumas, 2022)
Replications with confidence intervals	yes	no; one run per simulation, no replication setting documented	no; one run per call, repeatable only by an external script
Common random numbers across scenarios	yes, per-source streams	not documented; engine not published	no per-source streams; one global generator, seedable only from outside
Paired scenario comparison	yes	no	no
Value stream figures (VA/BVA/NVA, PCE)	yes, simulation-based	no	no
Licence	Apache-2.0	user interface MIT; engine not published	engine: no licence file; web services Apache-2.0

5. Discussion and limitations

The verification covers steady-state queueing behaviour, the semantics shared with Prosimos and the VSM partition; it does not cover every BPMN construct the engine executes. Event-based gateways, inclusive gateways, multi-instance activities and boundary events are covered by unit tests but not by an external reference; conformance tests against externally defined traces are future work. The resource model is coarser than that of Prosimos: there are no individual resource calendars, no resource-specific durations, no batching and no prioritisation, so models discovered by Simod have to be approximated (Section 4.3); the V3 result bounds but does not isolate the effect of these approximations. The tool reports confidence intervals and paired differences but leaves the choice among several significant options to the user; ranking-and-selection procedures (Kim & Nelson, 2001) would be a natural extension. The decision experiment (V5) uses two decision problems; it demonstrates the effect of single runs under controlled conditions and is not a general estimate of error rates. The VSM categories depend on the value classes assigned by the modeller, the precedence rule for overlapping work is a design choice, and the reported PCE measures elapsed-time coverage rather than work-content share (Sections 3.4, 4.4). The hash-seeded random streams are deterministic; their independence is assumed, and structural scenario changes can desynchronise paired draws (Section 3.3). No user study was conducted, so claims about usability are limited to the tool's availability in the browser and the modeller plugin.

6. Conclusion

The question that opened this paper now has a defensible answer: a single run per scenario with independent random numbers often picks the wrong option; common random numbers remove most of these errors, and a few paired replications with confidence intervals remove the rest. Process Simulator brings replicated, CRN-based scenario comparison and value stream figures for non-linear processes into a BPMN editor that runs in the browser and in the Camunda Modeler. The evaluation shows agreement with exact queueing results and with an independent engine, correct VSM figures for choices, parallel branches and loops, and a clear reduction of wrong decisions compared with single runs. The software, the experiments and the data are available under Apache-2.0.

Software and data availability

- Software (Apache-2.0):
  `https://github.com/BFH-Process-Digitalization/process-simulator`
  Web application:
  `https://bfh-process-digitalization.github.io/process-simulator/`
  The results were produced with release 1.0.1 (tag `v1.0.1`), archived on Zenodo:
  `https://doi.org/10.5281/zenodo.24219723`
- Experiments, generated models, raw runs and this manuscript:
  `https://github.com/BFH-Process-Digitalization/process-simulator-paper`
- The BPI Challenge 2012 log is not redistributed; the scripts download it from 4TU.ResearchData (van Dongen, 2012).

CRediT authorship contribution statement

Moritz Maier: Conceptualization, Software, Validation, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The author declares no competing interests.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used generative AI tools in order to draft and revise parts of the manuscript. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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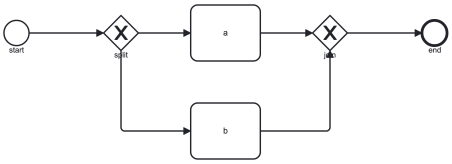
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Appendix A. Test models and decision scenarios

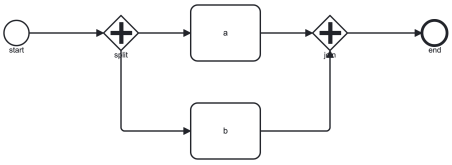
The BPMN models used in V1, V2, V4 and V5 are generated by the model builder in `experiments/lib/bpmn.mjs` and stored in the `models/` directory of the paper repository. Figure A.1 shows them as rendered by bpmn-js; the sequence of three activities of V4 has the same structure as the office process of Figure A.2, which Section 4.5 compares under the five scenarios of Table A.1. Figure A.3 shows the coffee-shop example shipped with the tool, whose stored scenarios Table A.2 lists as extracted from the model file. The BPI Challenge 2012 model of Section 4.3 is discovered by Simod 5.2.1 from the event log and regenerated by `make v3`; it is too large to reproduce legibly here.



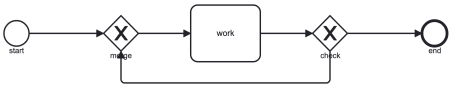
(a) single station (V1)



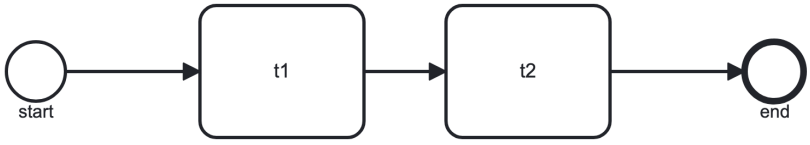
(b) exclusive choice (V2, V4)



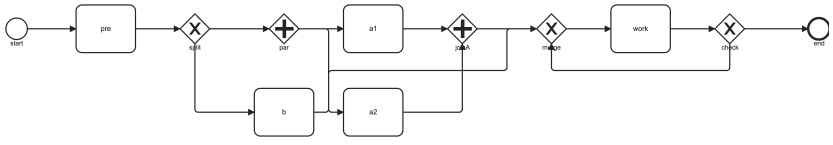
(c) parallel split (V2, V4)



(d) rework loop (V1, V2, V4)



(e) tandem of two stations (V1, V2, V4)



(f) choice + parallel pair + rework loop (V4)

Figure A.1: Generated BPMN test models used in the experiments; sources in `models/`. The rework loop (d) is also the Jackson feedback network of V1.



Figure A.2: The office process of the V5 decision experiment: three steps in sequence (task ids `t1`, `t2`, `t3`, served by roles `r1`, `r2`, `r3`); the same model builder produces the three-activity sequence of V4.

Table A.1: Office-process scenarios of V5 (Figure A.2). Each run simulates 2000 cases with exponential inter-arrival times of mean 60 s and exponential activity durations (means in seconds); staff counts give the number of resources of roles `r1`, `r2` and `r3` serving steps 1, 2 and 3.

Option	Step 1 (s)	Step 2 (s)	Step 3 (s)	Staff 1/2/3
Baseline	48	100	45	1/2/1
A: second person at step 1	48	100	45	2/2/1
B: step 2 10 % faster	48	90	45	1/2/1
C: step 3 11 % faster	48	100	40	1/2/1
D: second person at step 3	48	100	45	1/2/2

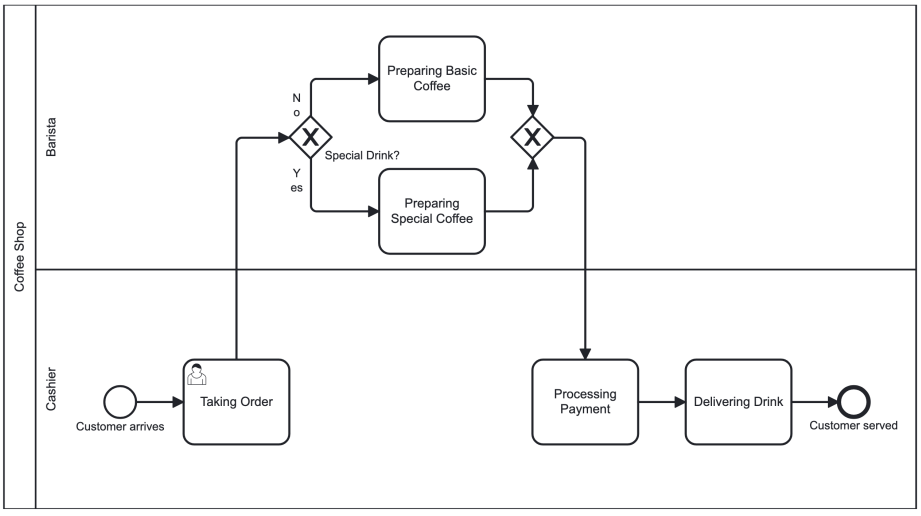


Figure A.3: The coffee-shop example shipped with the tool, used in the decision experiment (V5); Prosimos could not run it (Section 4.2).

Table A.2: Coffee-shop scenarios stored in the model file of Figure A.3, extracted from the embedded scenario block; all times in seconds. The variants list only the parameters they override.

Baseline	
Arrival	exponential, mean 300 s (one customer every 5 min)
Cases and horizon	at most 200 cases or a horizon of 28 800 s (8 h)
Start date, currency, seed	2026-11-16, CHF, 42
Resources	Cashier 1 and Barista 1 (30 CHF/h each, 24/7)
Taking Order	normal 120 ± 30 s, Cashier; BVA, queue limit 5
Processing Payment	fixed 60 s, Cashier; BVA
Delivering Drink	normal 30 ± 10 s, Cashier; NVA
Preparing Basic Coffee	normal 180 ± 60 s, Barista; VA, revenue 4
Preparing Special Coffee	normal 300 ± 120 s, Barista; VA, revenue 6
Gateway "Special Drink?"	"Yes" (special) 0.3, "No" (basic) 0.7
Variants (changes only)	
Customers every 3 min	arrival mean 180 s
Second barista	Barista count 2
Product mix 50/50	branch probabilities 0.5 / 0.5
Every 3 min + second barista	arrival mean 180 s; Barista count 2